Vinayak Kamat

Python Backend Engineer | AI/ML & LLM Infrastructure

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Summary

Python-focused AI/ML Engineer at KAN Innovations with experience building and deploying production AI systems, LLM applications, and backend services. Hands-on experience with FastAPI, RAG pipelines, LLM response analysis, API integrations, and AWS infrastructure. Built Aethelgard, a FastAPI-based LLM governance gateway for real-time response risk analysis, and optimized ML inference using PyTorch and ONNX. Interested in building reliable, scalable AI infrastructure and developer tools.

Skills

Programming: Python, C++, JavaScript, Dart

Backend & APIs: FastAPI, Flask, REST APIs, API integration, Docker

LLM & AI Infrastructure: LLM APIs, RAG, LangChain, LlamaIndex, llama.cpp, GGUF, Chroma, prompt engineering

ML & Deep Learning: PyTorch, Transformers, OpenCV, spaCy, ONNX

Cloud & MLOps: AWS Lambda, S3, ECR, DynamoDB, MLflow, model versioning, experiment tracking

Data & Computing: NumPy, Pandas, Linux

Experience

KAN Innovations, Junior AI/ML Engineer
Mumbai, Maharashtra
May 2025 – present
1 year 7 months

- Built and evaluated Retrieval-Augmented Generation (RAG) pipelines for AI-powered applications.
- Integrated machine learning models into production services and deployment workflows using AWS Lambda, ECR, S3, and DynamoDB.
- Optimized model inference by converting PyTorch models to ONNX, improving runtime performance.
- Improved model calibration to enhance the reliability of production ML predictions.
- Used MLflow for experiment tracking, model versioning, and reproducible ML workflows.

KAN Innovations, ML Engineer Intern
Mumbai, Maharashtra
Feb 2025 – May 2025
3 months

- Developed a PyTorch-based ulcer detection model achieving approximately 85% accuracy.
- Improved the performance of an eye posture detection model through model development and optimization.

Projects

Aethelgard | LLM Governance Gateway
FastAPI-based AI governance gateway for real-time LLM response analysis, hallucination detection, and risk assessment.

- Built an LLM response analysis pipeline using a secondary model for semantic risk assessment.
- Implemented a FastAPI gateway to process and analyze LLM responses.
- Developed real-time risk metric streaming and Datadog integration for monitoring.
- Designed automated hallucination and response-risk detection logic to improve the reliability of AI applications.

nanoMind | Offline AI Assistant
On-device AI chat application running language models locally using llama.cpp and Kotlin.
- Integrated GGUF language models for fully offline inference on Android devices.
- Implemented real-time token streaming using Kotlin Coroutines and Jetpack Compose.
- Optimized model loading and inference for ARM-based devices.

Financial AI Agent
Python-based prototype for financial data retrieval and reasoning using language models.
- Built retrieval and reasoning pipelines for financial information.
- Implemented modular components for query routing and response refinement.

Education

MSc **WorldQuant University**, Financial Engineering — Remote
Sept 2025 – May 2027

B.Tech **Punjab Technical University**, Computer Science — Punjab, India
Sept 2021 – May 2025

Achievements

Qualified the Graduate Aptitude Test in Engineering (GATE).

Participated in Kaggle competitions focused on applied machine learning.

Active open-source contributor with AI/ML projects on GitHub.